

Aurelio Amerio

PHD CANDIDATE

Valencia, Spain

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Areas of Expertise: Generative AI, Inverse Design problems, Surrogate Models, Diffusion & Flow Models, Multimodal Learning, LLMs and Agents, Bayesian Inference, Computational Physics.

Education

PhD

UNIVERSITY OF VALENCIA, INSTITUTO DE FISICA CORPUSCOLAR

PhD project on theoretical astroparticle physics and machine learning methodologies applied to physics.

Research advisor: Dr Bryan Zaldívar

Valencia, Spain

Feb 2022 – Current

Master's degree

UNIVERSITY OF TORINO

Thesis title: "Machine learning approach to the determination of the gamma-ray source count distribution below the Fermi-LAT detection limit"

Research advisor: Prof. Nicolao Fornengo and Prof. Alessandro Cuoco

Final grade: **110/110**

Turin, Italy

2018 – 2021

Violin diploma

CONSERVATORIO A. VIVALDI

Music diploma, equivalent to a master's degree

Alessandria, Italy

2007 – 2018

Bachelor's degree

UNIVERSITY OF TORINO

Thesis topic: gravitation lensing

Research advisor: Prof. Nicolao Fornengo

Final grade: **108/110**

Turin, Italy

2014 – 2017

Programming experience (4 years)

Selected Open Source Software

GenSBI: An open-source JAX library for **Implicit Likelihood problems**, implementing state-of-the-art generative models (**Flow Matching, Diffusion, Normalizing Flows, 6k+ downloads**).

Concrete Dropout: Published a popular implementation for Bayesian uncertainty estimation in Tensorflow/PyTorch (**35k+ downloads**).

MultiQuad.jl: A Julia package for high-performance n-dimensional numerical integration.

Scientific Machine Learning & Generative AI

Focus: Simulation-Based Inference (**SBI**), Neural Density Estimation (Continuous Normalizing Flows, **Diffusion Models**), Differentiable Programming, Uncertainty Quantification, Surrogate Modeling, and Probabilistic Programming.

Core Stack: **JAX** (Flax, Optax, Diffax), **PyTorch**, NumPyro, Blackjax.

Research Interest: **Agentic AI** for autonomous discovery, implicit likelihood methods for inverse design problems, physics informed machine learning.

High-Performance Computing & Numerical Simulation

Focus: GPU-accelerated computing, Differentiable Physics, and numerical solvers for ODEs/SDEs.

Languages: **Julia** (Expert), **Python** (Expert), C++, Mathematica.

Funded projects

Center for Computational Astrophysics

FUNDED BY: SIMONS FOUNDATION

New York, NY (USA)

Nov. 2025 - Jan 2026

During this 3-month research stay, I collaborated with Prof. Mariangela Lisanti, Prof. Francisco Villaescusa-Navarro, and Dr. Elena Pinetti focusing on axion detection, dark matter filaments, and Simulation-Based Inference applications. This project was supported by a **\$15.000** grant to cover accommodation and living expenses.

Laboratoire d'Annecy-le-Vieux de Physique Théorique

FUNDED BY: UNIVERSITY OF VALENCIA

Annecy, France

28 Apr. - 28 May 2025

This project involved a 1 month research stay under the supervision of prof. Pasquale Serpico and prof. Francesca Calore. This project was awarded **1800€** in funds.

Fermilab

FUNDED BY: HORIZON EUROPE PROGRAMME (ASYMMETRY GRANT)

Chicago, IL (USA)

Mar. - Apr. 2024

This project involved a 2-month research stay at Fermilab (USA), where I had the opportunity to work with the astroparticle theory group under the supervision of Prof. Dan Hooper.

This project was awarded **4600€** in funds.

Novel methods in Dark Matter searches driven by Artificial Intelligence

FUNDED BY: CONSELLERIA DE EDUCACIÓN, INVESTIGACIÓN, CULTURA Y DEPORTE DE LA COMUNITAT VALENCIANA

Valencia, Spain

2022 - Current

This is the scholarship funding my doctoral studies, for a total of **53.000€** over three years.

Visiting scientist

Flatiron Institute & New York University

CENTER FOR COMPUTATIONAL ASTROPHYSICS

New York, NY (USA)

Nov. 2025 - Jan. 2026

Flatiron Institute

CENTER FOR COMPUTATIONAL ASTROPHYSICS

New York, NY (USA)

24 Mar. - 9 Apr. 2025

Publications

GenSBI: Generative Methods for Simulation-Based Inference in JAX

A. AMERIO

2026

Software, arXiv:**2605.27499**

A JAX library (6k+ downloads) implementing state-of-the-art generative models for simulation-based inference, including flow-matching and diffusion models, as well as normalizing flows for general-purpose (conditional) neural density estimation.

Search for dark matter subhalos among Fermi-LAT sources in presence of dataset shift

A. AMERIO, D. MALYSHEV, B. ZALDÍVAR, V. GAMMALDI, M. A. SÁNCHEZ-CONDE

2025

Accepted by JCAP, arXiv:**2503.14584**

Dark satellites are among the most compelling targets for dark matter annihilation searches in our galaxy. I use machine learning to probe the unassociated gamma-ray sources detected by Fermi-LAT for a population of dark matter subhalos, explicitly accounting for the dataset shift between training simulations and real data.

Dark Matter and Galaxy Cross-Correlations with the Cherenkov Telescope Array Observatory

E. PINETTI, V. VODEB, A. AMERIO, A. CUOCO, S. CAMERA, N. FORNENGO, G. ZAHARIJAS

2025

Phys. Rev. D 112 (2025) 12, 123010, arXiv:**2505.20383**

I quantify the prospects of the CTA extragalactic survey for detecting a cross-correlation between the observed gamma-ray emission and tracers of standard astrophysical sources, such as active galactic nuclei, and assess the resulting sensitivity to a dark matter signal.

Millisecond Pulsars in Globular Clusters and Implications for the Galactic Center Gamma-Ray Excess

A. AMERIO, D. HOOPER, T. LINDEN

2024

JCAP10(2025)106, arXiv:**2412.05220**

I derive the gamma-ray luminosity function of millisecond pulsars in globular clusters and find it consistent with measurements of pulsars along the galactic plane. I then use this result to test whether a millisecond-pulsar population can account for the galactic center excess.

A. AMERIO, F. CALORE, P. D. SERPICO, B. ZALDÍVAR

JCAP03(2024)055, arXiv:**2306.16483**

I propose a statistical method to extend Fermi-LAT catalogs of high-latitude gamma-ray sources below their nominal threshold. By simulating ensembles of synthetic skies, I convert the purely statistical information of the source-count distribution into a per-pixel likelihood that low test-statistic regions host a genuine source. The method supports multi-wavelength follow-up of new source candidates and is particularly valuable for cross-correlation analyses.

A. AMERIO, A. CUOCO, N. FORNENGO

JCAP09(2023)029, arXiv:**2302.01947**

I reconstruct the extragalactic gamma-ray source-count distribution (dN/dS) of resolved and unresolved sources with deep learning, training a convolutional neural network on synthetic sky-maps that span a range of source-count models and include the Fermi-LAT instrument response. Applied to 14 years of Fermi-LAT data in the 1–10 GeV band, the network recovers dN/dS down to fluxes a factor of 50 below the detection threshold, matching catalog-derived results in the resolved regime and extending as S^{-2} down to $5 \cdot 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$.

Invited seminars and conference contributions

Columbia University THEORETICAL HIGH ENERGY ASTROPHYSICS GROUP Seminar	New York, NY (USA) 23 Jan. 2026
Center for Computational Astrophysics FLATIRON INSTITUTE - PACMAN SEMINAR Seminar	New York, NY (USA) 7 Nov. 2025
Center for Computational Astrophysics FLATIRON INSTITUTE - COSMOLOGY GROUP MEETING Seminar	New York, NY (USA) 26 Mar. 2025
New York University CENTER FOR COSMOLOGY AND PARTICLE PHYSICS Seminar	New York, NY (USA) 31 Mar. 2025
University of Turin PHYSICS DEPARTMENT Seminar	Turin, Italy 22 Jan. 2025
IDM2024 UNIVERSITY OF L'AQUILA, INFN, GSSI Poster	L'Aquila, Italy 8-12 July 2024
Invisibles school 2024 UNIVERSITY OF BOLOGNA, INFN Poster	Bologna, Italy 24-28 June 2024
Fermilab MACHINE LEARNING SEMINAR SESSIONS Chalk talk	Chicago, IL (USA) 26 Apr. 2024
Fermilab ASTROPARTICLE THEORY SEMINAR SESSIONS Invited Seminar	Chicago, IL (USA) 25 Mar. 2024
European Astronomical Society annual meeting 2022 EUROPEAN ASTRONOMICAL SOCIETY (EAS) Parallel session talk	Valencia, Spain June 2022

Workshops and schools

Towards realistic detection forecasts of primordial GWBs

UNIVERSITY OF VALENCIA

Workshop

Valencia, Spain

9-12 Dec. 2024

European Consortium for Astroparticle Theory School

EuCAPT AND UNIVERSITY OF VALENCIA

One-week postgraduate school

Valencia, Spain

Sep. 2023

Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation

INFN, GGI, AND UNIVERSITY OF FLORENCE

Three-week postgraduate school

Florence, Italy

Apr. 2024

Taller de Altas Energías 2022

CENTRO DE CIENCIAS DE BENASQUE PEDRO PASCUAL

Two-week workshop on high energy physics and astrophysics

Benasque, Spain

Sep. 2022

Open Science and Outreach

Machine Learning journal club

ORGANIZER @ UNIVERSITY OF VALENCIA

A journal club with the aim to discuss and teach the basics of modern ML techniques to PhD students.

2023-2025

From Zero to Julia!

FOUNDER AND WRITER

I created a comprehensive lecture series on **Julia for scientists**, recognized as a **top learning resource** by the official Julia language community and cited in university curricula.

2021-Current

Manning

EXTERNAL REVIEWER

Subject matter expert and technical reviewer for Julia publications.

2022-Current

Reference contacts

Bryan Zaldívar

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Dan Hooper

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Nicolao Fornengo

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Dmitry Malyshev

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August 14, 2026